Exploring the Role of Network Metrics in Infectious Disease Transmission: A Case Study on Medical Contact Networks

Giverphine Dejiga
College of Information and Computing
University of Southeastern Philippines
Davao City, Philippines
gdejiga01487@usep.edu.ph

Matthew Feri Tanutan
College of Information and Computing
University of Southeastern Philippines
Davao City, Philippines
mftanutan00220@usep.edu.ph

Hans C. Toledo
College of Information and Computing
University of Southeastern Philippines
Davao City, Philippines
hctoledo00115@usep.edu.ph

Abstract

Hospitals are high-risk environments for infectious disease transmission due to dense and frequent human interactions. This study explored the structural dynamics of a hospital contact network using social network analysis to identify individuals with high potential for disease spread, by applying centrality metrics, specifically PageRank, Eigenvector, Degree, and Betweenness. The researchers assessed the influence of healthcare workers, patients, and staff within the network. Community detection algorithms, including Leiden and Girvan–Newman, examined the organization's internal structure. The analysis revealed a cohesive, single-community structure, suggesting tightly integrated hospital operations. It also showed that nurses consistently held the highest centrality scores across all measures, highlighting their central role in hospital interactions and their potential as key spreaders during outbreaks. Diffusion simulations using a fixed transmission probability demonstrated that infections originating from high-centrality individuals led to faster and more widespread transmission. In contrast, low-centrality individuals exhibited delayed and limited infection spread. The findings of the study provide insights into prioritization of infection control and support network-based strategies in hospital outbreak management.

Keywords: Social Network Analysis, Hospital Contact Network, Centrality Measures, Infectious Disease Transmission, Diffusion

Introduction

Infectious diseases are the greatest threats to organizations' ability and productivity, particularly in structured environments such as hospitals, schools, and workplaces. Therefore, given such conditions, a single infected person can easily spread the infection to many others, causing a major outbreak in the organization. Due to the recent outbreak that spread worldwide, the coronavirus SARS-CoV-2 highlighted the importance of contact among humans in the rapid spread of the virus infection. By then, the pandemic had been declared by the World Health Organization (WHO). In just a couple of months, the number scaled to millions of infections worldwide and caught nations red-handed as it put considerable stress on health care systems and economies worldwide [1]. Because of this, it became increasingly clear that understanding how the virus spreads through human interactions was essential. These interactions act as invisible highways for viral transmission, particularly in dense environments like hospitals, workplaces, and schools. Understanding how diseases move through these networks is essential for designing effective responses, such as quarantine measures, contact tracing, or targeted vaccinations.

A recent study [2] conducted a comprehensive literature review in analyzing 1,752 publications from 2001 to 2020 exclusively on the interdisciplinary study of social network analysis and the prevention and control of epidemics. The study explores how social networks have evolved over time and argues that understanding different network structures is key to effectively managing infectious disease outbreaks. It provides valuable insights into how diseases propagate through social networks, highlighting key factors of influence and evaluating various prevention and control strategies. The study analyzes patterns of interactions among

individuals and pinpoints critical nodes and connections that could be targeted to mitigate the transmission of diseases.

A study [3] conducted pioneering research in a geriatric hospital ward using a wearable RFID proximity sensor for real-time recording of contact events among health care workers (HCWs), patients, and administrative staff. The findings showed that a tiny fraction of HCWs, especially nurses, contributed to a disproportionately high segment of total patient contacts, identifying them as potential super-spreaders in the chains of hospital-acquired infections. The evidence from this can empirically support network analysis in infection control and demonstrates how heterogeneous contact patterns, role-based channeling, and time-dependent contact dynamics can alter the spread of disease. These insights underscore the importance of investigating disease transmission within hospital settings to better inform targeted control measures.

Research Objectives

This study aims to investigate disease transmission within a hospital setting by analyzing the structure and dynamics of the hospital contact network. Specifically, it seeks to address the following research questions through social network analysis and simulation:

1. Which individuals in the hospital contact network are identified as central figures using centrality measures such as eigenvector centrality, PageRank, degree, and betweenness centrality?
2. How do central individuals influence the speed and extent of disease transmission in simulated diffusion processes across the medical contact network?

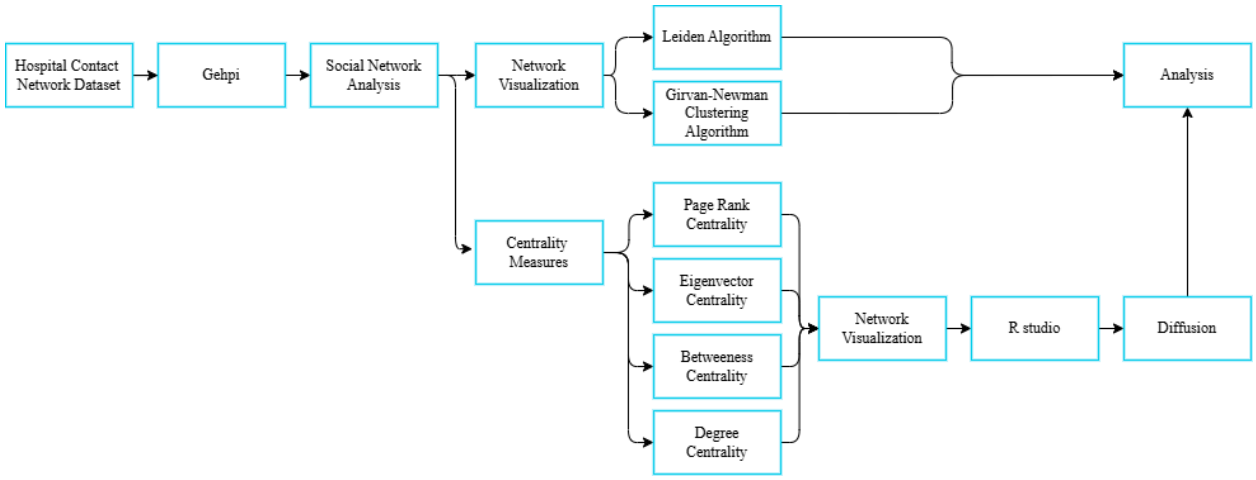
3. What are the observed differences in diffusion patterns when an infection is initiated by high-centrality versus low-centrality individuals?

Conceptual Framework

Hospital contact networks offer a crucial lens through which disease transmission patterns can be explored and understood. In this study, the researchers used the SocioPatterns Hospital Ward dataset [3], which captures face-to-face interactions between healthcare workers (HCWs) and patients over a four-day period. Figure 1 shows the conceptual framework for this study, illustrating the workflow from data collection to network analysis and infection diffusion modeling. Each phase contributes to understanding the role of contact structure in facilitating or mitigating disease transmission in a confined healthcare setting.

Figure 1.

Social Network Analysis Conceptual Framework



Materials and Methods

Data Collection

In this study, the researchers used the SocioPatterns Hospital Ward dataset, which contains a temporal network of contacts between individuals in a healthcare setting [3]. The dataset was collected from the SocioPatterns collaboration (www.sociopatterns.org), a platform that collects and shares high-resolution contact data for studying human interactions. In the referenced study, researchers used IoT-enabled devices to detect close-range face-to-face interactions among hospital staff and patients. The dataset only covers a four-day period, from the afternoon of Monday, December 6, 2010, to the afternoon of Friday, December 10, 2010, in a hospital ward in Lyon, France, and involved a total of 75 participants, including 46 healthcare workers (HCWs) and 29 patients.

The dataset captures contact events as a tab-separated list, where interactions between individuals are recorded in 20-second intervals throughout the observation period. Each record includes a timestamp (t), two anonymized individual IDs (i and j), and their corresponding roles within the hospital setting (Si and Sj). The roles are categorized as 'NUR' for paramedical staff (including nurses and aides), 'PAT' for patients, 'MED' for medical doctors, and 'ADM' for administrative staff.

Preprocessing

The raw dataset was initially converted into a CSV format to facilitate compatibility with Gephi, the software researchers used for network analysis. Preprocessing was conducted within the Gephi environment, wherein the dataset was restructured to generate two primary

components: a node list and an edge list. The node list contained unique individual identifiers and their corresponding roles, while the edge list captured the source–target contact pairs and their associated timestamps. This formatting enabled a structured dataset representation that was suitable for subsequent visualization and centrality analysis. The dataset was then exported in R as a GraphML file for diffusion modeling, allowing for further simulation of spread dynamics across the network.

Social Network Analysis

Social Network Analysis (SNA) is an explicit methodology designed to analyze patterns of interaction between people or things, which represent themselves as nodes (actors) in a network connected by edges (relationships) [4]. This approach allows researchers to measure and interpret the structural properties of social systems, particularly regarding how information, behaviors, or diseases can be spread within nodes through common connections. SNA has been applied quite extensively within epidemiology to assess the transmission dynamics of infectious diseases through mapping contact networks and identifying individuals who might act as super-spreaders.

In this study, SNA serves as a valuable tool for gaining insights into hospital contact patterns by analyzing interactions between health care workers, patients, and staff. This analysis helps identify which departments or roles are most at risk, providing a clearer understanding of potential transmission pathways. By applying centrality measures such as eigenvector centrality and PageRank and by applying the community detection algorithms Girvan-Newman and Leiden, researchers can analyze the structural dynamics of the network and simulate potential

disease diffusion. Through this approach, SNA enables researchers to gain meaningful insights into the network's behavior and identify key individuals or groups for analysis.

Centrality Measurements

Centrality measures were used in this study to help analyze the hospital contact network by identifying key individuals who may influence the spread and control of infections. Table 1 summarizes these measures and their significance.

Table 1.

Summary of Centrality Measurements

Centrality Measurement	Brief Description	Significance
PageRank	A centrality measure that ranks the importance of nodes based on the number and quality of their connections [5]	Helps identify the most influential nodes in a network, allowing researchers to understand which nodes most strongly affect the network's structure and behavior [5].
Eigenvector Centrality	Assigns relative scores to all nodes in the network based on the principle that connections to high-scoring nodes contribute more to the score of the node [6].	Identifies not just well-connected nodes but also those connected to other well-connected nodes, revealing influential actors embedded in important substructures [6].
Degree Centrality	Quantifies the number of direct connections (edges) a node has in	Highlights individuals with the most direct interactions, identifying them

	a network [7].	as highly exposed or influential in spreading infection within the network [7].
Betweenness Centrality	Identifies nodes that act as bridges in the shortest paths between other nodes [8].	Highlights individuals who control information flow and connect different parts of the network[8].

Community Detection

To explore how individuals naturally form groups within the hospital contact network, this study applies methods for identifying communities in the dataset. Table 2 outlines the algorithms used and their significance.

Table 2.

Summary of Community Detection Algorithms

Algorithm	Brief Description	Significance
Leiden Algorithm	An improved version of Louvain that guarantees well-connected, accurate communities while optimizing modularity. It is fast, scalable, and avoids disconnected clusters [9].	It guarantees well-connected communities, improved accuracy and stability, modularity optimization in terms of partition, and it is computationally faster and more efficient than Louvain [9].
Girvan-Newman Clustering Algorithm	Identifies communities by progressively removing edges with the highest betweenness centrality. It is effective but computationally slow for large networks [10].	It enables it to work well on small to medium networks and reveals its hierarchical structure based on structural properties [10].

Diffusion

Diffusion in the social network refers to the spread of information in the interconnected network of nodes and their entities [11]. It depends on the rate and intensity of diffusion on its topology and the network parameters' initialization. In this study, the researchers simulated diffusion processes through Hospital Contact Networks to better understand how infectious diseases propagate within real healthcare environments. Diffusion modeling is a means to simulate the spread of pathogens through an interaction between persons recorded in the network structure. To carry out the simulation, the researchers utilized the RStudio environment, leveraging R-based packages and scripts to model the spread of infection across the network.

In a realistic modeling of the infection process, the proponents have a transmission probability diffusion rate of 0.1, which indicates that in any contact event, an infected node has a 10% chance of passing the infection to a connected neighbor. This conservative rate is meant to reflect the understanding that not all close-range encounters lead immediately to infection transmission. The effect of diffusion process modeling sought to understand how infections could travel along key pathways in the network and how certain individuals could accelerate or inhibit disease propagation. It examines comprehensively how infections can propagate through intricate human interaction networks and how the network structure affects the result of transmission.

Tools

In this study, the researchers utilized a set of tools to support the visualization, analysis, and interpretation of data. These tools were selected based on their capabilities in generating clear and meaningful visual representations, such as graphs, charts, and network diagrams, which are essential for understanding data relationships and patterns. Table 3 outlines the specific tools used for visualization and analysis throughout the research process.

Table 3.

Overview of Visualization and Statistical Tools

Tools	Brief Description
Gephi	Gephi is the leading visualization tool due to its open-source and free nature. It is a visualization application developed in Java mainly for visualizing, manipulating, and exploring networks and graphs from raw edge and node graph data [12]. In this study, it was used for network visualization, allowing the proponents to map and analyze the relationships and connections within the dataset effectively. Gephi's capabilities in rendering complex network structures and supporting various layout algorithms make it ideal for identifying patterns and insights within the data. Researchers used the Gephi 0.10.1 version in this study.
R Studio	RStudio is an integrated development environment (IDE) specifically designed for R programming. It offers a comprehensive and user-friendly platform for data analysis, statistical computing, and visualization [13]. In this study, researchers used R Studio to generate diffusion models and graphical representations that supported the overall analysis. The version utilized in this study is R programming 12.1, released in 2024.

Results and Discussion

Table 4 summarizes the key topological characteristics of the network analyzed in this study. The graph was constructed and visualized using the Fruchterman-Reingold layout algorithm with specified parameters for area and gravity for better visualization.

Table 4.

Summary of Graph Topological Metrics

Layout Parameters	2500 area; 60 gravity
Nodes	75
Edges	32424
Graph Density	11.684
Average Degree	864.4

Figure 2.

Communities Based Leiden Algorithm Sized By Eigenvector Centrality

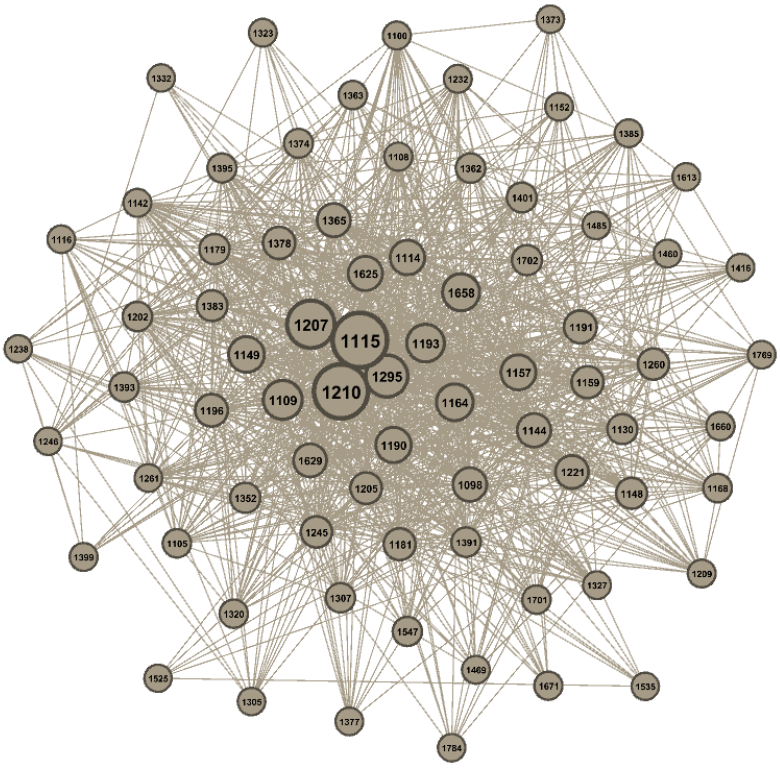


Figure 2 shows that the Leiden Clustering Algorithm detected only one community within the Hospital Contact Network, indicating a closely connected and unified network. This suggests that the network functions as a single organizational unit, likely reflecting the integrated nature of hospital operations where staff and patients frequently interact across departments. As a result, the researchers treated the entire network as one organization for analysis. The varying node sizes are based on eigenvector centrality, highlighting individuals who are not only central but also well-connected to other influential nodes, potentially key personnel such as nurses or doctors with high mobility.

Figure 3.

Communities Based Girvan Newman Algorithm Sized By Eigenvector Centrality

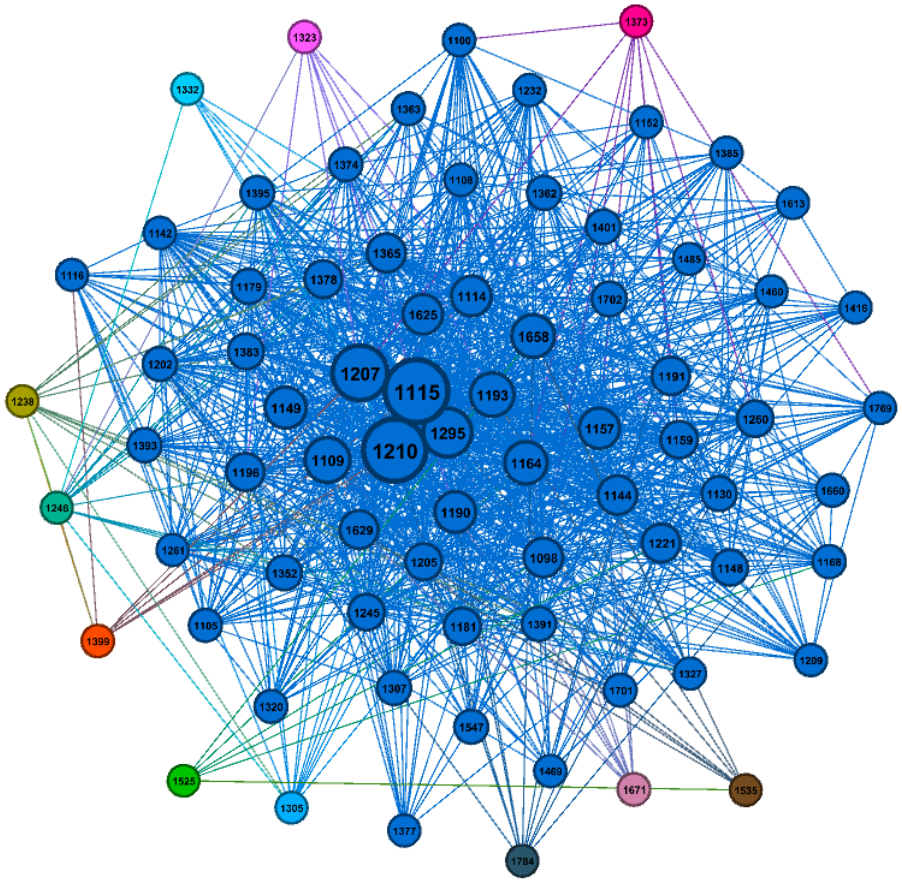


Figure 3 shows the communities detected through the Girvan–Newman algorithm, with the node size representing eigenvector centrality. To compare with the previous result from the Leiden algorithm, which identified only one community, this model was applied to explore whether a more granular community structure could be detected. While the Girvan–Newman algorithm does reveal multiple color-coded groups, these minor clusters represent nodes with fewer interactions and are likely statistical outliers rather than distinct functional communities. This algorithm tends to split the network by progressively removing the most "between" edges, which can result in isolating nodes on the network's fringe that have limited ties to the densely connected core. Despite these minor separations, the majority of nodes remain within a dominant blue-colored community, suggesting a still cohesive and unified network structure. Hence, the researchers concluded that, even under a different algorithm, the network essentially behaves as a single community, with the few colored outliers reflecting low-interaction individuals rather than meaningful subdivisions.

Figure 4.

Hospital Contact Network Based on PageRank Centrality

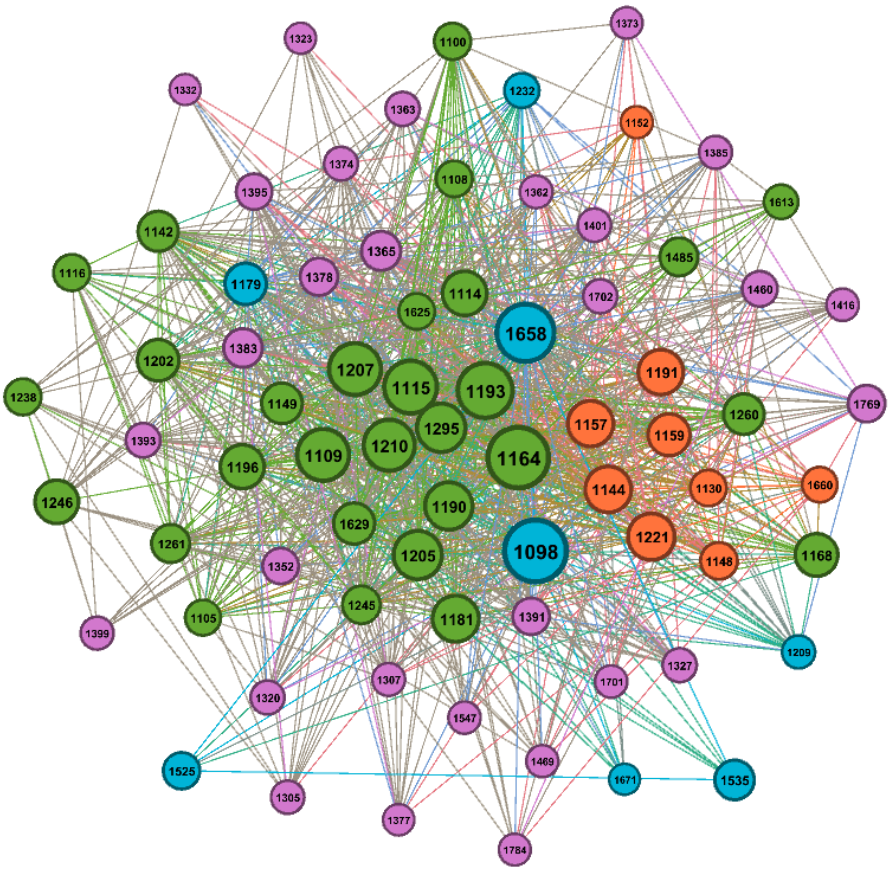


Table 5.

Highest PageRank Centrality of the Hospital Contact Network

Id	Role	Pagerank
1098	Admin	0.005365
1164	Nurse	0.005234
1658	Admin	0.004883
1193	Nurse	0.004613
1115	Nurse	0.004392
…	…	…

Table 6.

Lowest PageRank Centrality of the Hospital Contact Network

Id	Role	PageRank
1332	Patient	0.002081
1671	Admin	0.002082
1323	Patient	0.002097
1377	Patient	0.002126
1373	Patient	0.002147
…	…	…

Figure 4 shows the network visualization of the Hospital Contact Network based on PageRank centrality, where node colors indicate different roles within the hospital. Specifically, pink represents patients, green indicates nurses and paramedic aides, orange corresponds to medical doctors, and blue represents administrators. Each node corresponds to an individual in the hospital, while the edges represent direct contact between them. The size of each node reflects its PageRank value, highlighting the most influential individuals in the network. The visualization underscores the structural importance of different roles: individuals with high PageRank scores, primarily administrators and nurses, play a crucial role in facilitating contact and the flow of information. In contrast, patients generally have lower PageRank values, suggesting they have fewer connections and limited influence within the overall network structure.

Figure 5.

Hospital Contact Network Based on Eigenvector Centrality

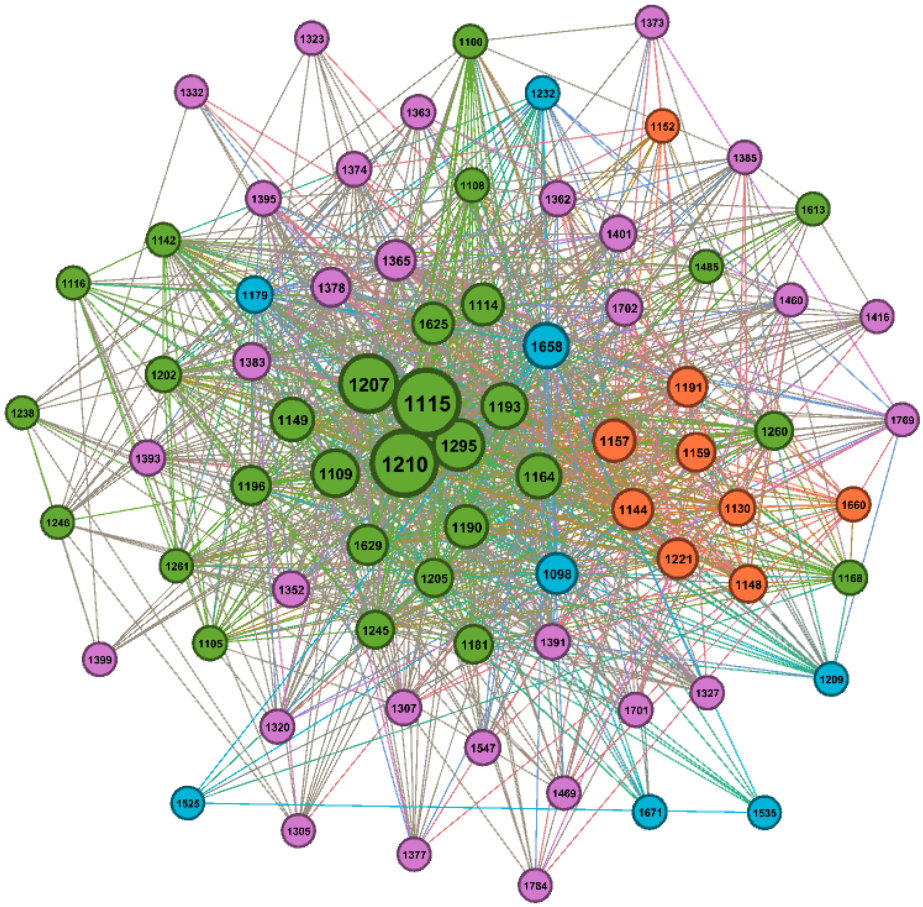


Table 4.

Highest Eigenvector Centrality of the Hospital Contact Network

Id	Role	Eigenvector
1115	Nurse	1
1210	Nurse	0.9818
1207	Nurse	0.7413
1295	Nurse	0.5355
1109	Nurse	0.4279
…	…	…

Table 5.

Lowest Eigenvector Centrality of the Hospital Contact Network

Id	Role	Eigenvector
1525	Admin	0.000954
1535	Admin	0.002915
1238	Nurse	0.003312
1332	Patient	0.003615
1784	Patient	0.006077
…	…	…

Figure 5 represents the network visualization of the Hospital Contact Network now based on Eigenvector Centrality, with node colors corresponding to the roles as described in Figure 4. The size of each node reflects its eigenvector centrality score, which is a measure of the significance of a node, not just based on connections but also based on the connections of its neighboring nodes. Thus, higher eigenvector centrality identifies individuals who are central to the most influential parts of the network.

The most prominent nodes in this visualization are nurses, indicating that they occupy central structural positions and maintain strong connections with other well-connected individuals. Therefore, it shows the nursing staff's important role in terms of the interaction dynamics within the hospital. On the contrary, the smallest nodes are the least significant in the network, with the lowest eigenvector centrality scores, representing individuals who are minimally connected and not linked to central figures in the network. These individuals are positioned on the periphery of the network and have limited impact on its overall structure.

From a real-world perspective, individuals with high eigenvector centrality, such as nurses, in this case, can be key drivers in the spread of infectious diseases if exposed, due to their frequent and close contact with other central individuals. Their embedded position in the network means that an infection reaching them could quickly propagate through the core of the hospital network, making it crucial to prioritize infection control measures for these highly connected staff members.

Figure 6.

Hospital Contact Network Based on Degree Centrality

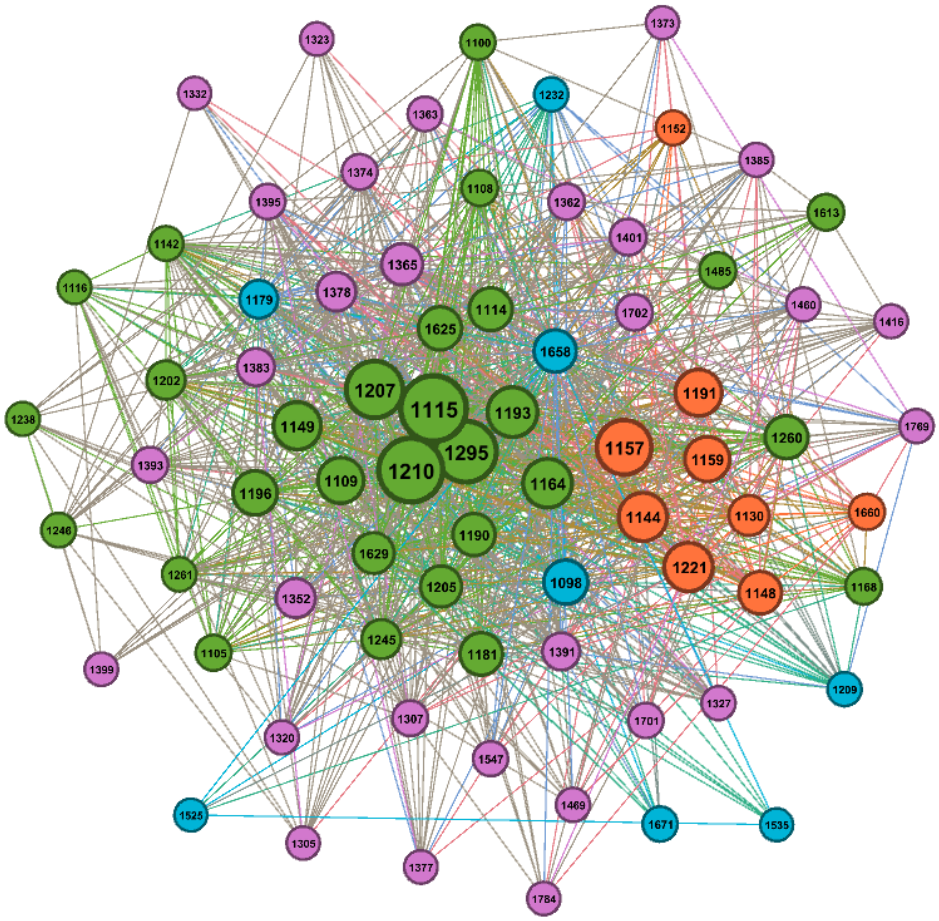


Table 6.

Highest Degree Centrality of the Hospital Contact Network

Id	Role	Degree
1115	Nurse	4286
1210	Nurse	4077
1295	Nurse	3695
1207	Nurse	3130
1157	Medical	2849
…	…	…

Table 7.

Lowest Degree Centrality of the Hospital Contact Network

Id	Role	Degree
1525	Admin	12
1535	Admin	18
1332	Patient	21
1373	Patient	46
1784	Patient	61
…	…	…

Figure 6 represents the network visualization of the Hospital Contact Network based on Degree Centrality, with node colors corresponding to the roles as described in Figure 4. Degree centrality implies the number of direct connections a node has with other nodes in the network. This shows that most of those with the highest number of contacts are nurses; thus, these are the core of the system and are highly influential in the flow of interactions across the system. The

administrative and patient have the lowest degree of centrality, implying that they are at the network's periphery, with very few interactions.

Figure 7.

Hospital Contact Network Based on Betweenness Centrality

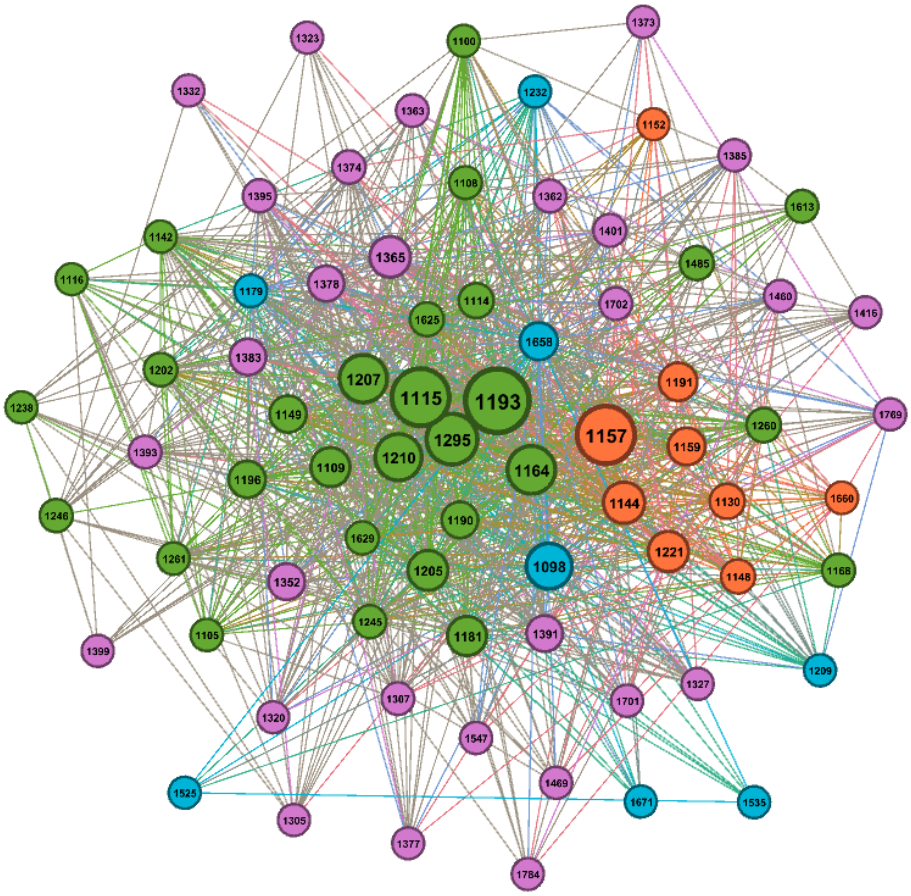


Table 8.

Highest Betweenness Centrality of the Hospital Contact Network

Id	Role	Betweenness
1193	Nurse	180.29
1157	Medical	145.27
1115	Nurse	144.23

1295	Nurse	107.23
1164	Nurse	90.46
…	…	…

Table 9.

Lowest Betweenness Centrality of the Hospital Contact Network

Id	Role	Betweenness
1671	Admin	0.005558
1373	Patient	0.015608
1377	Patient	0.022652
1332	Patient	0.026839
1323	Patient	0.103376
…	…	…

Figure 7 shows the Hospital Contact Network based on betweenness centrality, which measures how often an individual connects others through the shortest paths in the network, highlighting key intermediaries in communication flow. Node size reflects this centrality, with larger nodes playing more crucial roles. As shown in Table 8, nurses consistently hold the highest betweenness scores, placing them at the core of the network and emphasizing their role as essential connectors within the hospital. In contrast, Table 9 and the smaller, more peripheral nodes, such as patients and admin staff, indicate limited involvement in bridging interactions. This underscores the central coordinating role nurses play in maintaining the hospital's communication structure.

Diffusion

To investigate the role of network position in disease transmission, researchers simulated diffusion processes initiated by individuals with the highest and lowest scores across various centrality measures. A transmission rate of 0.1 was used to represent the probability of infection per contact. The simulation aimed to observe how the infection propagated through the network over time and to determine the number of days required to infect all nodes.. The rate of infection over time was visualized using a line graph generated in R, illustrating the rate and speed of spread for each centrality scenario. The following results summarize the outcomes of these simulations.

Figure 8.

Example of Diffusion in Network Visualization as initiated by the Highest Degree Centrality

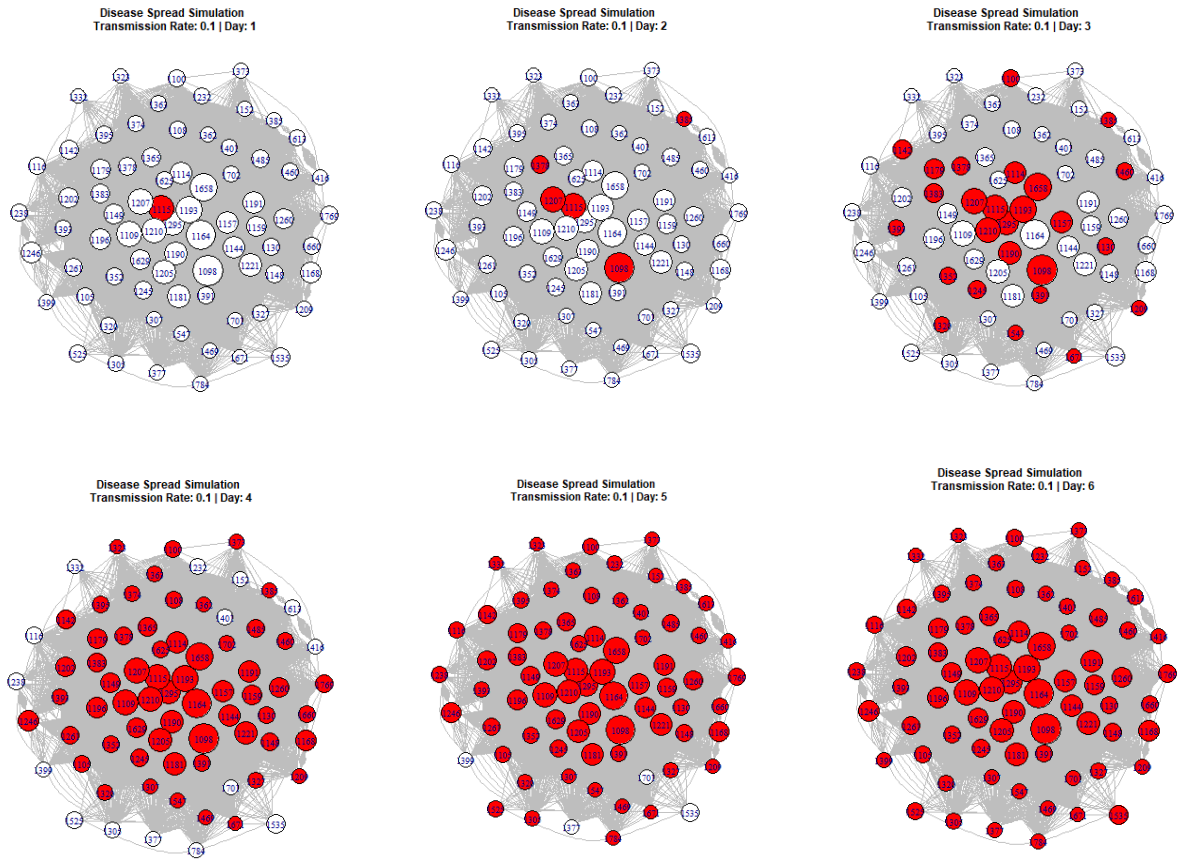
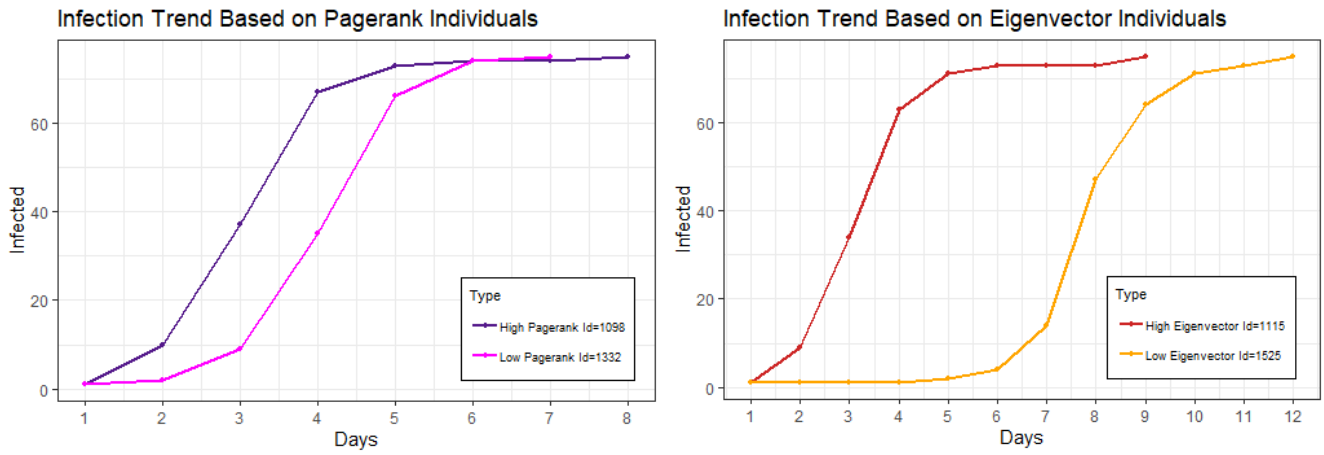


Figure 8 illustrates an example of diffusion in the network, initiated by the individual with the highest degree centrality (Node ID: 1115). This visualization simulates the spread of a disease at a transmission rate of 0.1 over six days.

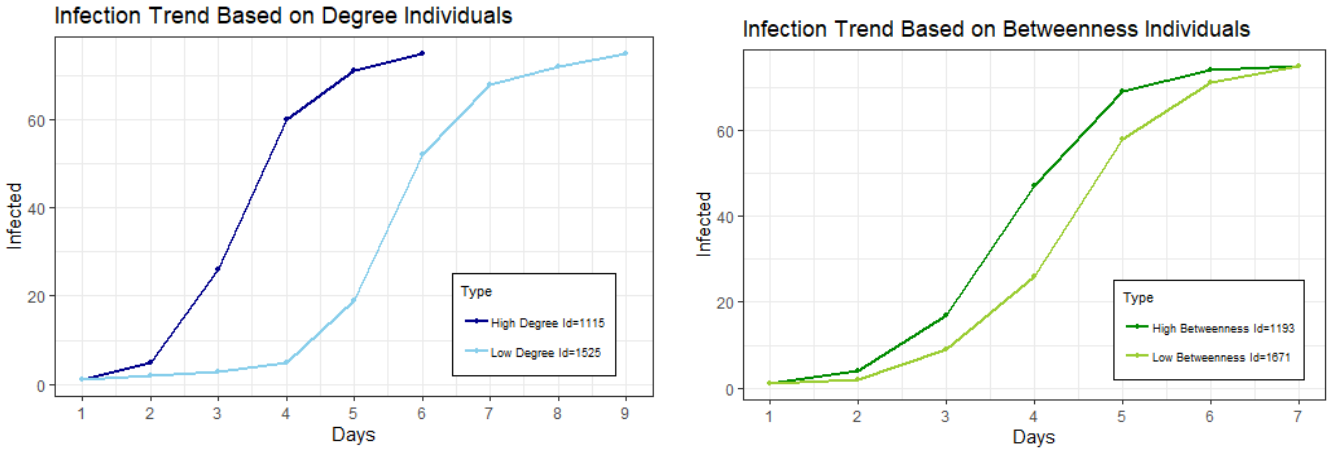
Figure 9.

Infection Initiated by Highest and Lowest Centrality Individuals in Line Graph



(a) PageRank

(b) Eigenvector Centrality



(c) Degree Centrality

(d) Betweenness Centrality

The diffusion results shown in Figure 9 reveal a clear pattern across all centrality measures: individuals with the highest centrality scores tend to spread infection faster and more effectively than those with the lowest scores. For example, in the PageRank (Figure 9a) and Degree Centrality (Figure 9c) graphs, the infection spreads rapidly when initiated by high-centrality individuals, with a noticeable surge by Day 2, infecting more than half of the network by Day 4. In contrast, the spread starts more slowly when initiated by low-centrality individuals, with the increase in infections only becoming significant around Day 4. This shows that individuals with more direct contacts (degree) or stronger influence within the network (PageRank) are more capable of quickly reaching others.

A similar pattern is seen in Eigenvector Centrality (Figure 9b). Here, the high-centrality individual causes a sharp rise in infections starting on Day 2, while the low-centrality counterpart only begins to spread the infection around Day 7 significantly. Since eigenvector centrality measures how well an individual is connected to other well-connected individuals, this result suggests that such individuals are highly effective in initiating early diffusion. In Betweenness Centrality (Figure 9d), both the high and low centrality individuals eventually lead to full infection at roughly the same final day, but the speed of spread still varies. The high betweenness individual starts the spread earlier in Day 2, while the low betweenness individual only triggers faster transmission around Day 4. This shows that while final outcomes may align, the timing of the infection burst depends on the individual's position in the network. Overall, these findings confirm that individuals with higher centrality scores play a critical role in early-stage transmission, helping to spread infection faster than others. These insights highlight how network structure and individual position influence the timing and reach of disease spread, which is valuable in designing effective prevention and intervention strategies.

CONCLUSION

This study highlights the significant potential of social network analysis in understanding the transmission dynamics of infectious diseases within hospital settings. By applying various centrality measures, including PageRank, eigenvector, degree, and betweenness centrality, this study identified individuals, particularly nurses, as central figures within the contact network. Their roles, marked by frequent and widespread interactions, place them at the core of the hospital's structural connectivity, positioning them as potential super-spreaders in the event of an outbreak. Community detection results further revealed that the hospital operates as a single, tightly connected community, suggesting a high level of interaction across roles and departments. This unified structure contributes to the rapid diffusion of infection when initiated by high-centrality individuals, as shown in the simulation outcomes.

The diffusion simulations confirmed that individuals with higher centrality scores significantly influence the speed and reach of disease spread. In contrast, individuals with lower centrality were less effective at propagating infection across the network. This reinforces the importance of identifying and potentially prioritizing central individuals, such as nurses, in infection control strategies, particularly during early outbreak stages.

However, the study also presents important limitations that must be acknowledged. The contact dataset was limited in scope, capturing only four days of interactions in a single ward and lacking crucial epidemiological variables such as infection status, incubation period, and individual susceptibility. Moreover, the model employed a constant transmission probability, which oversimplifies the complex, variable nature of disease transmission. Critically, this study

did not account for the temporal order of contacts, when each interaction occurred, which is essential for accurately modeling how an infection spreads over time.

To overcome these limitations, future research should incorporate temporal network analysis and utilize larger, more diverse datasets across departments or institutions. Temporal data would allow researchers to capture the progression of contact events, enabling more realistic simulations of disease transmission and enhancing outbreak prediction accuracy. Incorporating variables such as infection status and individual risk profiles would further refine these models. Ultimately, these improvements could lead to the development of dynamic, real-time intervention strategies tailored to the evolving structure of contact networks, improving both preparedness and response in healthcare environments.

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